



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

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Laboratory Manual

Academic Year	:	2026-27	Semester	:	3 rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical – 7

PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

Problem Statement:- Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages.

Objective:- To develop a PHP-based server-side form processing system that validates and sanitizes user inputs, securely stores submitted registration/contact data in CSV or JSON format, and displays appropriate success or error messages.

Github Repository Link:- <https://github.com/Nishtha-cse/StudentHubPortal-D26CS129>

index.php :

```
<?php
session_start();

$errors = $_SESSION['errors'] ?? [];
$oldData = $_SESSION['old_data'] ?? [];
$success = $_SESSION['success'] ?? '';

unset($_SESSION['errors'], $_SESSION['old_data'], $_SESSION['success']);

function old(array $data, string $key): string {
    return htmlspecialchars($data[$key] ?? '', ENT_QUOTES, 'UTF-8');
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Registration</title>

    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&di
splay=swap" rel="stylesheet">
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <main class="form-card">
        <h1>Create your registration</h1>
        <p class="intro">Fill in your details below. All fields are
required.</p>

        <?php if ($success): ?>
            <div class="alert success">
                <?= htmlspecialchars($success) ?>
            </div>
        <?php endif; ?>

        <?php if ($errors): ?>
            <div class="alert error" role="alert">
                <strong>⚠ Please correct the following errors:</strong>
                <ul>
                    <?php foreach ($errors as $error): ?>
                        <li><?= htmlspecialchars($error) ?></li>
                    <?php endforeach; ?>
                </ul>
            </div>
        <?php endif; ?>
    </main>
</body>
</html>
```

```

        </div>
    <?php endif; ?>

    <!-- novalidate lets the server-side errors above appear instead of
    browser popups -->
    <form action="process.php" method="POST" novalidate>
        <div class="form-grid">

            <div class="field">
                <label for="name">Name</label>
                <input type="text" id="name" name="name"
autocomplete="name"
                    value="<? = old($oldData, 'name') ?>">
            </div>

            <div class="field">
                <label for="email">Email</label>
                <input type="email" id="email" name="email"
autocomplete="email"
                    value="<? = old($oldData, 'email') ?>">
            </div>

            <div class="field">
                <label for="mobile">Mobile</label>
                <input type="text" id="mobile" name="mobile"
inputmode="numeric"
                    maxlength="10" autocomplete="tel-national"
                    value="<? = old($oldData, 'mobile') ?>">
            </div>

            <div class="field">
                <label for="pincode">Pincode</label>
                <input type="text" id="pincode" name="pincode"
inputmode="numeric"
                    maxlength="6" autocomplete="postal-code"
                    value="<? = old($oldData, 'pincode') ?>">
            </div>

            <fieldset class="field full">
                <legend>Gender</legend>
                <div class="radio-row">
                    <input type="radio" id="male" name="gender"
value="male"
                        <? = ($oldData['gender'] ?? '') === 'male' ?
'checked' : '' ?>>
                    <label for="male">Male</label>

```

```

        <input type="radio" id="female" name="gender"
value="female"
        <?= ($oldData['gender'] ?? '') === 'female' ?
'checked' : '' ?>>
        <label for="female">Female</label>
    </div>
</fieldset>

    <div class="field full">
        <label for="address">Address</label>
        <input type="text" id="address" name="address"
autocomplete="street-address"
        value="<?= old($oldData, 'address') ?>">
    </div>

    <div class="field full">
        <label for="message">Message</label>
        <textarea id="message" name="message"><?= old($oldData,
'message') ?></textarea>
    </div>

</div>

    <div class="actions">
        <button type="submit">Submit</button>
    </div>
</form>
</main>
</body>
</html>

```

Process.php :

```

<?php
session_start();

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    header('Location: index.php');
    exit();
}

$storage_mode = 'json'; // 'json' or 'csv'

/* ----- Read input ----- */
$name      = trim($_POST['name'] ?? '');
$email     = trim($_POST['email'] ?? '');
$mobile    = trim($_POST['mobile'] ?? '');
$pincode   = trim($_POST['pincode'] ?? '');

```

```
$gender = trim($_POST['gender'] ?? '');
$address = trim($_POST['address'] ?? '');
$message = trim($_POST['message'] ?? '');

// Kept so the form can be refilled if validation fails
$oldData = compact('name', 'email', 'mobile', 'pincode', 'gender', 'address',
'message');

/* ----- Validate ----- */
$errors = [];

if ($name === '') {
    $errors[] = 'Name is required.';
} elseif (!preg_match('/^[a-zA-Z ]{2,50}$/', $name)) {
    $errors[] = 'Name must contain only letters and spaces.';
}

if ($email === '') {
    $errors[] = 'Email address is required.';
} elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    $errors[] = 'Enter a valid email address.';
}

if ($mobile === '') {
    $errors[] = 'Mobile number is required.';
} elseif (!preg_match('/^[0-9]{10}$/', $mobile)) {
    $errors[] = 'Mobile number must contain exactly 10 digits.';
}

if ($pincode === '') {
    $errors[] = 'Pincode is required.';
} elseif (!preg_match('/^[0-9]{6}$/', $pincode)) {
    $errors[] = 'Pincode must contain exactly 6 digits.';
}

if (!in_array($gender, ['male', 'female'], true)) {
    $errors[] = 'Please select a valid gender.';
}

if ($address === '') {
    $errors[] = 'Address is required.';
}

if ($message === '') {
    $errors[] = 'Message is required.';
}

if ($errors) {
```

```
$_SESSION['errors'] = $errors;
$_SESSION['old_data'] = $oldData;
header('Location: index.php');
exit();
}

/* ----- Sanitize ----- */
$entry = [
    'timestamp' => date('Y-m-d H:i:s'),
    'name'      => htmlspecialchars($name, ENT_QUOTES, 'UTF-8'),
    'email'     => htmlspecialchars($email, ENT_QUOTES, 'UTF-8'),
    'mobile'    => htmlspecialchars($mobile, ENT_QUOTES, 'UTF-8'),
    'pincode'   => htmlspecialchars($pincode, ENT_QUOTES, 'UTF-8'),
    'gender'    => htmlspecialchars($gender, ENT_QUOTES, 'UTF-8'),
    'address'   => htmlspecialchars($address, ENT_QUOTES, 'UTF-8'),
    'message'   => htmlspecialchars($message, ENT_QUOTES, 'UTF-8'),
];

/* ----- Store ----- */
function saveToCSV(string $file, array $entry): bool
{
    $isNew = !file_exists($file);
    $handle = fopen($file, 'a');
    if (!$handle || !flock($handle, LOCK_EX)) {
        return false;
    }
    if ($isNew) {
        fputcsv($handle, array_keys($entry));
    }
    fputcsv($handle, array_values($entry));
    flock($handle, LOCK_UN);
    fclose($handle);
    return true;
}

function saveToJSON(string $file, array $entry): bool
{
    $records = file_exists($file) ? json_decode(file_get_contents($file),
true) : [];
    if (!is_array($records)) {
        $records = [];
    }
    $records[] = $entry;
    return file_put_contents($file, json_encode($records, JSON_PRETTY_PRINT),
LOCK_EX) !== false;
}

if ($storage_mode === 'csv') {
```

```
$saved = saveToCSV('data.csv', $entry);
$where = 'CSV';
} else {
    $saved = saveToJSON('data.json', $entry);
    $where = 'JSON';
}

if ($saved) {
    $_SESSION['success'] = "Registration successful! Data stored in $where
file.";
} else {
    $_SESSION['errors'] = ['Unable to save registration.'];
    $_SESSION['old_data'] = $oldData;
}

header('Location: index.php');
exit();
```

Registration Form :

Create your registration

Fill in your details below. All fields are required.

Name	Email
Mobile	Pincode
Gender	
Male	Female
Address	
Message	
Submit	

Required Field Validation :

Create your registration

Fill in your details below. All fields are required.

⚠ Please correct the following errors:

- Name is required.
- Email address is required.
- Mobile number is required.
- Pincode is required.
- Please select a valid gender.
- Address is required.
- Message is required.

Name

Email

Mobile

Pincode

Gender

Male

Female

Address

Message

Submit

Name Format Validation :

Create your registration

Fill in your details below. All fields are required.

⚠ Please correct the following errors:

- Name must contain only letters and spaces.
- Email address is required.
- Mobile number is required.
- Pincode is required.
- Please select a valid gender.
- Address is required.
- Message is required.

Name

Email

Mobile

Pincode

Gender

☐ Male

☐ Female

Address

Message

Submit

Email Format Validation :

Create your registration

Fill in your details below. All fields are required.

⚠ Please correct the following errors:

- Name is required.
- Enter a valid email address. Email must be a valid address like name@example.com, with a real domain, no spaces, and at most 100 characters.
- Mobile number is required.
- Pincode is required.
- Please select a valid gender.
- Address is required.
- Message is required.

Name

Email

Mobile

Pincode

Gender

Male

Female

Address

Message

Submit

Mobile Number Format Validation :

Create your registration

Fill in your details below. All fields are required.

⚠ Please correct the following errors:

- Name is required.
- Mobile number must contain exactly 10 digits.
- Pincode is required.
- Please select a valid gender.
- Address is required.
- Message is required.

Name

Email

Mobile

Pincode

Gender

Male

Female

Address

Message

Submit

Gender Selection Validation:

Create your registration

Fill in your details below. All fields are required.

⚠ Please correct the following errors:

- Please select a valid gender.
- Address is required.
- Message is required.
- You must accept the terms and conditions.

Name	Email
nishtha	nishtha@gmail.com
Mobile	Pincode
1234567890	390016
Gender	
Male	Female
Address	
Message	
☐ I agree to the terms and conditions	
Submit	

Terms & Conditions Validation :

Create your registration

Fill in your details below. All fields are required.

⚠ Please correct the following errors:

- You must accept the terms and conditions.

Name	Email
nishtha	nishtha@gmail.com
Mobile	Pincode
1234567890	390016
Gender	
Male	Female
Address	
sdfghjkl	
Message	
asdfghjkl	
☐ I agree to the terms and conditions	
Submit	

Registration Successful :

Create your registration

Fill in your details below. All fields are required.

Registration successful!

Name	Email
Mobile	Pincode
Gender	
Male	Female
Address	
Message	
☐ I agree to the terms and conditions	
Submit	

Data Stored in CSV File successfully :

```
data.csv
1 "2026-09-28 16:41:21",nishtha,nishtha@gmail.com,1234567890,123456,f
2
```

Data Stored in Json File successfully :

```
{ } data.json > ...  
1  [  
2      {  
3          "timestamp": "2026-09-28 16:33:18",  
4          "name": "nishtha",  
5          "email": "nishtha@gmail.com",  
6          "mobile": "1234567890",  
7          "pincode": "390016",  
8          "gender": "female",  
9          "address": "sdfghjkl",  
10         "message": "asdfghjkl"  
11     },  
12     {  
13         "timestamp": "2026-09-28 16:34:15",  
14         "name": "nishtha",  
15         "email": "nishtha@gmail.com",  
16         "mobile": "1234567890",  
17         "pincode": "123456",  
18         "gender": "female",  
19         "address": "qwertyuio",  
20         "message": "qwertyuiop"  
21     }  
22  ]
```